Kids 2D Vector Animation Web App

Requirements for Claude Code

Target User: Children aged 6–10 | **Platform:** Web Browser (Chrome, Firefox, Safari, Edge)

---

1. Overview

A fun, easy-to-use 2D vector animation web app designed for kids. The interface should feel playful and approachable — big buttons, bright colors, simple icons, and clear labels. Think of it like a kid-friendly browser-based version of Adobe Animate, inspired by the colorful cartoon style seen on channels like Florzy Studio on YouTube.

The app lives at a URL, requires login, and works on any device with a modern browser.

---

2. Tech Stack Recommendation

- **Frontend:** React + Vite
- **Rendering:** Fabric.js (canvas-based 2D vector manipulation)
- **Animation Engine:** Custom keyframe system with GSAP for previewing
- **Video Export:** FFmpeg.wasm (runs FFmpeg entirely in the browser, no server needed)
- **Auth:** Supabase Auth (username/email + password, remember me)
- **Database & Storage:** Supabase (project saves, asset storage)
- **Hosting:** Vercel or Netlify (free tier)
- **Language:** TypeScript + React

---

3. SEO & Crawler Settings

3.1 Robots Meta Tag

- Add the following to the root `index.html` and all app pages:

```html
<meta name="robots" content="noindex, nofollow">
```

- This prevents search engines from indexing the app or following any links

3.2 robots.txt

- Include a `robots.txt` file at the root of the domain:

```
User-agent: *
Disallow: /
```

- Blocks all crawlers from the entire app

3.3 Additional Headers

- Set `X-Robots-Tag: noindex, nofollow` in server response headers
- Ensures crawlers that ignore meta tags are also blocked

---

4. Authentication System

4.1 Login Page

- Clean, kid-friendly login screen with the app logo and mascot character
- Fields:
  - **Email or Username**
  - **Password** (show/hide toggle)
  - **Remember Me** checkbox (persists session for 30 days via secure cookie)
- **Login** button (large, colorful)
- Link to **Sign Up** page
- Link to **Forgot Password** (sends reset email)
- No social login (keep it simple)

4.2 Sign Up Page

- Fields:
  - **Username** (display name, letters/numbers only)
  - **Email** (parent's email recommended)

    - **Password**
    - **Confirm Password**
- Password requirements: minimum 8 characters
- **Create Account** button
- Link back to Login

4.3 Session Management

- If **Remember Me** is checked: session persists 30 days
- If **Remember Me** is unchecked: session expires when browser is closed
- Secure HttpOnly cookie for session token
- Auto-redirect to login if session expires mid-use
- Logout button always accessible from top bar

4.4 Security

- Rate limit login attempts (max 5 per minute)
- HTTPS only
- Passwords hashed (handled by Supabase Auth)
- No sensitive data stored in localStorage

4.5 Protected Routes

- All app pages require authentication
- Unauthenticated users are always redirected to `/login`
- Only public pages: `/login`, `/signup`, `/forgot-password`

---

5. UI & Design Requirements

5.1 General Feel

- Bright, colorful, cartoon-style interface
- Large clickable buttons with icons AND text labels
- Tooltips on hover for every tool
- Undo/Redo always visible (Ctrl+Z / Ctrl+Y)
- No confusing menus — everything visible or one click away

- Dark and Light mode toggle
- Fully responsive for tablet and desktop (minimum 768px width)

5.2 Layout

```
┌──────────────────────────────────────────────────────────────────────┐
│ [Logo]  [New] [Open] [Save]   [Username] [Logout] 🎬 │ ← Top Bar
├──────────┬──────────────────────────────────────────┬──────────────┤
│          │                  │           │
│ TOOLS    │     CANVAS / STAGE        │ PROPERTIES  │
│ PANEL    │                   │ PANEL       │
│ (left)   │                   │ (right)     │
│          │                  │           │
├──────────┴──────────────────────────────────────────┴──────────────┤
│          TIMELINE / KEYFRAME EDITOR              │
└──────────────────────────────────────────────────────────────────────┘
```

5.3 Home / Dashboard Screen (after login)

- Greeting: "Hi [Username]! Ready to animate? 🎨"
- Grid of saved projects with thumbnail previews
- **+ New Project** button (large, prominent)
- Recent projects (last 5)
- Achievement badges earned so far

---

6. Tools Panel (Left Sidebar)

6.1 Drawing & Shape Tools

- **Select Tool** — click and drag to select/move objects
- **Rectangle Tool** — draw squares and rectangles
- **Circle/Ellipse Tool** — draw circles and ovals
- **Triangle Tool** — draw triangles
- **Star Tool** — draw stars (3–12 points, adjustable)
- **Polygon Tool** — draw custom polygons
- **Line Tool** — draw straight lines
- **Pencil/Freehand Tool** — draw freehand paths (converts to vector)

- **Text Tool** — add text with kid-friendly font options

6.2 Edit Tools

- **Fill Color Picker** — solid color or gradient
- **Stroke Color Picker** — outline color and thickness
- **Eraser Tool** — remove parts of shapes
- **Group/Ungroup** — combine shapes into one object
- **Duplicate** — copy any object instantly
- **Delete** — remove selected object

---

7. Canvas / Stage

- Default canvas size: **1920x1080 (16:9, YouTube standard)**
- Optional sizes: 1280x720, 1080x1080 (square)
- Zoom in/out (mouse scroll or +/- buttons)
- Drag canvas to pan when zoomed in
- Rulers and optional snap-to-grid
- Background color picker for the stage
- Multiple **scenes** — kids can chain multiple scenes per project

---

8. Properties Panel (Right Sidebar)

When an object is selected, show:

- **Position** — X, Y (simple sliders)
- **Size** — Width, Height with lock aspect ratio toggle
- **Rotation** — 0–360 degree dial or input
- **Opacity** — 0–100% slider
- **Fill** — solid color, gradient, or texture
- **Stroke** — color, thickness, dashed options
- **Layer Order** — move forward/backward buttons
- **Name** — let kids name their objects (e.g. "Happy Star")

---

9. Color & Texture System

9.1 Color

- Large color picker with:
  - Preset palette of 30+ bright kid-friendly colors
  - Custom color wheel
  - Hex input
  - Eyedropper tool
- Recent colors bar

9.2 Gradients

- Linear and radial gradient support
- Simple 2-color gradient editor

9.3 Textures & Patterns

- Built-in library of fun textures:
  - Dots, stripes, zigzag, stars, checkerboard
  - Grass, clouds, brick, wood, fur, scales
- Apply textures as fills to any shape
- Import a custom image as a texture/fill

---

10. Import System

10.1 Supported Import Formats

- **SVG —** import vector images (characters, backgrounds, props)
- **PNG / JPG —** import bitmap images
- **GIF —** import animated GIFs as reference

10.2 Built-in Asset Library

- Pre-made character parts (heads, eyes, mouths, arms, legs)

- Background templates (sky, space, underwater, city, forest)
- Props (trees, clouds, stars, vehicles, animals)
- All assets SVG-based and colorable after import

10.3 Import Behavior

- Drag and drop files directly onto the canvas
- Browse button in toolbar
- Imported SVGs are editable (colors changeable)
- Auto-scale to fit canvas on import
- Assets stored in Supabase Storage per user account

---

11. Animation System

11.1 Timeline (Bottom Panel)

- Horizontal timeline with frames/seconds ruler
- Each object appears as its own layer row
- Scrubber bar to preview any frame
- Play / Pause / Stop controls
- Frame rate: 12, 24, or 30 FPS
- Total duration up to 10 minutes

11.2 Keyframe Animation

- Right-click any object on timeline to **Add Keyframe**
- Drag keyframes to change timing
- Auto-interpolation (tweening) between keyframes for:
    - Position (move across screen)
    - Scale (grow/shrink)
    - Rotation (spin)
    - Opacity (fade in/out)
    - Color (color change over time)
- Visual easing presets: **Bounce, Smooth, Snap, Elastic**

11.3 Simple Animation Presets

One-click animation effects:

- 🎉 **Pop In** — scales from 0 to full size
- 🌊 **Wiggle** — side-to-side wobble
- 💫 **Spin** — continuous rotation
- ⬆️ **Float Up** — rises from bottom
- 👋 **Wave** — arm-wave motion
- ✨ **Sparkle** — opacity flicker
- 🚀 **Fly In** — slides in from any direction
- 🔄 **Loop** toggle — make any animation repeat

11.4 Scene Transitions

- Fade in/out between scenes
- Slide left/right between scenes
- Instant cut

---

12. Audio (Phase 2)

- Import MP3/WAV background music
- Audio track on timeline
- Volume control
- Trim audio to match video length

---

13. Export System

13.1 Export to Video

- Format: **MP4 (H.264)** via FFmpeg.wasm — runs in browser, no server needed
- Resolution: 720p or 1080p
- Frame rate matches project setting
- Export progress bar with fun animation (rocket launching 🚀)

- Download triggers browser's native file save dialog

13.2 Export to Image

- Export current frame as PNG
- Export all frames as PNG sequence (zip download)

13.3 Export to GIF

- Export short clips as animated GIF (max 15 seconds)

13.4 Save Project

- Save to Supabase (cloud) — always available on any device
- Auto-save every 3 minutes with visible indicator
- Manual save button (Ctrl+S)
- Project stored as JSON with all layers, keyframes, and asset references

---

14. Kid-Friendly UX Features

- **Welcome Screen** — "What do you want to make today?" with templates
- **Tutorial Mode** — guided first project (a bouncing ball)
- **Sticker Mode** — drag-and-drop stickers/emojis onto canvas
- **Undo history** — minimum 50 steps
- **Big friendly error messages** — no technical jargon
- **Achievement badges** — "First Animation!", "Color Master!", "Exported!"
- **Character mascot** — guide character in corner giving tips
- **Autosave indicator** — always visible

---

15. Performance Requirements

- Initial page load under 3 seconds
- Smooth canvas interaction at 60fps
- Handle up to 100 objects without lag
- FFmpeg.wasm loaded lazily (only when export is triggered)

- Works on 4GB RAM devices
- Chrome, Firefox, Safari, Edge — latest 2 versions supported

---

16. Development Phases

Phase 1 — Core (MVP)

- Login / Signup / Remember Me / Logout
- noindex/nofollow on all pages
- Canvas with shape tools and color system
- Basic timeline with keyframes (position, scale, rotation)
- Import SVG/PNG
- Cloud save to Supabase
- Export MP4 via FFmpeg.wasm

Phase 2 — Polish

- Built-in asset library
- Animation presets
- Texture/pattern fills
- Scene transitions
- Tutorial mode
- Dashboard with project thumbnails

Phase 3 — Extras

- Audio support
- Achievement badge system
- Mascot guide character
- GIF export
- Parent dashboard (view child's creations)

---

17. Inspiration Reference

Style reference: Florzy Studio on YouTube — colorful, bold, character-driven 2D animations with smooth motion and expressive shapes. The app should make it easy for a child to recreate this style without any prior animation knowledge.